$U(1)$ Gauge Symmetry and the Origin of Electromagnetism

Khin Josmar V. Misuari*

Incoming Postgraduate Diploma Student, Abdus Salam
International Centre for Theoretical Physics (ICTP), Trieste, Italy

(Dated: May 31, 2026)

The $U(1)$ gauge formulation of electromagnetism is one of the most profound conceptual triumphs in modern physics, yet its elegance is frequently obscured by the dense mathematical machinery of standard quantum field theory textbooks. In this supplementary note, we provide a streamlined, physically motivated derivation of electromagnetism directly from symmetry principles without the technicalities inherent in gauge theories such as a thorough understanding of Lie groups. Starting with the free Dirac Lagrangian, we impose a single constraint: the freedom to alter the quantum phase of a wavefunction independently at every point in spacetime. As we will see, maintaining mathematical consistency under this local requirement inevitably dictates the introduction of the interacting gauge field A_μ, the field strength tensor $F_{\mu\nu}$, and ultimately, Maxwell's equations. We also explore the physical implications of gauge invariance, specifically detailing why it explicitly forbids a mass term for the gauge field A_μ. Exhaustive completeness is not the goal; rather, this note is designed to help interested students appreciate one of the true triumphs of modern physics and provide the foundational intuition needed to dive deeper into the gauge formalism.

To ensure that this resource remains highly accessible and self-contained, comprehensive appendices are included, which includes step-by-step derivation of the Dirac equation, a bottom-up construction of the Faraday tensor generalized from the classical Lorentz force, and an explicit derivation of Maxwell's equations in covariant form. This document bridges the gap between abstract gauge theory and observable classical electrodynamics, serving as a comprehensive guide for students preparing for advanced topics in theoretical physics such as the Standard Model. The reader is assumed to have a working familiarity with classical electromagnetism, quantum mechanics, special relativity, and the Lagrangian formulation of classical field theory.

* kj.misuari@gmail.com

I. INTRODUCTION

In modern physics, fundamental interactions are not arbitrarily patched together to fit experimental data; rather, they are mathematically mandated by symmetry (by that, we mean that the Lagrangian of a system remains invariant under a transformation of its fields, such as rotation, reflection, and so on). One of the biggest shifts in our understanding of the universe is the realization that symmetry is the fundamental concept, while interactions are merely its consequences. Each of the four fundamental interactions of nature is a direct, unavoidable consequence of a specific underlying symmetry requirement:

- **Gravity** arises from *diffeomorphism* symmetry (general covariance). This symmetry dictates that the laws of physics remain invariant under any arbitrary coordinate transformation, forcing us to abandon the notion of gravity as an external force and instead recognize it as the manifestation of curved spacetime geometry. In field theory, the unique nonlinear theory of interacting massless spin-2 gravitons is General Relativity (GR) [1, 2].

- **Electromagnetism** arises from local $U(1)$ gauge symmetry. As we will see, demanding that the quantum phase of a charged field can be shifted independently at any point in spacetime forces us to recover Maxwell's electromagnetism through the introduction of a compensating field, which manifests physically as the photon or light.

- The **weak interaction** arises from local $SU(2)$ gauge symmetry. This internal symmetry governs the flavor transformations of quarks and leptons, driving processes like nuclear beta decay. When unified with the $U(1)$ framework, it forms the electroweak interaction, a historic synthesis that ultimately predicted the existence of the Higgs boson particle [5].

- The **strong interaction** arises from local $SU(3)$ gauge symmetry. Operating under the framework of Quantum Chromodynamics (QCD), this symmetry acts on the "color charge" of quarks and gluons, generating the immense forces that permanently bind atomic nuclei together.

Together, the three symmetry groups (except gravity) comprise the gauge architecture of the Standard Model of particle physics: $SU(3) \times SU(2) \times U(1)$. The definitive discovery of the Higgs boson particle in 2012 [6] cemented the gauge formalism not merely as an elegant mathematical convenience, but as a foundational law of physical reality.

Electromagnetism is the most elegant and historically foundational example of the gauge formalism, arising entirely from local $U(1)$ gauge invariance. Here, the U stands for "unitary" (a property that preserves quantum probabilities), while the 1 indicates this is the simplest possible such group: its elements are ordinary complex numbers of magnitude 1, parameterized by a single angle α that is used to rotate the quantum field around the unit circle of the complex plane. Essentially, local $U(1)$ invariance means the phase of a charged field can be shifted independently at any point in spacetime. The elegance lies in how demanding this purely unphysical, local phase freedom naturally gives rise to one of the foundations of physics.

Undergraduate classical electromagnetism is fundamentally built upon two pillars: Maxwell's equations, which govern the fields, and the Lorentz force law, which governs the dynamics of particles [7]. Together, they are often presented as a collection of empirical rules that just happen to describe reality. The $U(1)$ gauge formulation flips this script, transforming electromagnetism into a mathematical necessity. However, standard quantum field theory (QFT) textbooks frequently bury this conceptual elegance under a mountain of index gymnastics, Feynman diagrams, and technical mathematical jargon. The goal of this document is to avoid these highly technical details as much as possible, serving as an accessible resource for undergraduate students.

The remainder of this document is organized as follows. In section II, we briefly review the foundations of classical field theory. In section III, we discuss Noether's theorem to establish the link between continuous symmetries and conserved currents. Section IV applies a global $U(1)$ transformation to the Dirac field to derive its conserved current. In section V, we promote this global symmetry to a local gauge transformation, mathematically motivating the introduction of a compensating gauge field A_μ. Section VI then demystifies the resulting field strength tensor by showing how it naturally recovers classical electric and magnetic fields alongside Maxwell's equations, while section VII explores the physical implications of gauge invariance, specifically detailing why it explicitly forbids a mass term for the gauge field. Finally, we summarize our conclusions in section VIII. Note that in this document, we will use the mostly negative metric signature $(+1, -1, -1, -1)$.

II. LAGRANGIAN FIELD THEORY AND THE DIRAC EQUATION

To ensure that this document remains as self-contained as possible, we briefly review the mathematical foundation required to build our gauge theory. In classical mechanics, the dynamics of a discrete point particle can be derived from the Lagrangian $L(t) = T - V$. But in field theory, the fields are continuously distributed across space and time, so we instead use the *Lagrangian density* $\mathcal{L}(\phi, \partial_\mu \phi)$, defined as:

$$L(t) = \int \mathcal{L}(\phi, \partial_\mu \phi)\, d^3x \tag{1}$$

Hence, the total action S of the system becomes an integral over all of spacetime:

$$S = \int L(t)\, dt = \iint \mathcal{L}(\phi, \partial_\mu \phi)\, d^3x\, dt \quad \Longrightarrow \quad S = \int \mathcal{L}(\phi, \partial_\mu \phi)\, d^4x \tag{2}$$

Hamilton's principle of least action ($\delta S = 0$) then determines the physical evolution of the system. Applying the calculus of variations to the action yields the Euler-Lagrange equation of motion for any given field ϕ:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = \frac{\partial \mathcal{L}}{\partial \phi} \tag{3}$$

For our purposes, the field of interest is the Dirac field ψ, a four-component spinor[1] that describes spin-1/2 fermions such as electrons. The governing relativistic wave equation for a free fermion is the Dirac equation:

$$(i\gamma^\mu \partial_\mu - m)\psi = 0 \tag{4}$$

where m is the mass of the particle, and γ^μ are the 4×4 Dirac matrices satisfying the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ (where $\eta^{\mu\nu}$ is the Minkowski metric). A detailed derivation of the Dirac equation can be found in Appendix A. To construct a scalar Lagrangian density that reproduces the Dirac equation via eq. (3), one can introduce the Dirac adjoint $\bar{\psi} \equiv \psi^\dagger \gamma^0$. Treating ψ and $\bar{\psi}$ as independent fields, the free Dirac Lagrangian density is written as:

$$\mathcal{L} = i\bar{\psi}\, \gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi \tag{5}$$

By "free" or "unperturbed", we simply mean that the particle is isolated and not interacting with any external forces, fields, or other particles[2]. It serves as the absolute baseline; physicists start with this mathematically simple, "switched-off" state before introducing interactions (or "perturbations"), such as an electromagnetic field, to see how the forces actually change the particle's behavior. Hence, this unperturbed Lagrangian density serves as our starting point. Our objective in the following sections is to determine exactly what happens to this free-particle system when we subject it to the rigid demands of quantum phase invariance.

[1] A spinor is essentially a complex mathematical object used to describe the intrinsic angular momentum (spin) of a fundamental particle.

[2] In other words, a "free Dirac Lagrangian" essentially describes a lonely electron coasting through a pure vacuum, much like a single spaceship drifting through deep space with its engines off and no gravity acting upon it.

III. SYMMETRY AND NOETHER'S THEOREM

In 1915, Emmy Noether (1882 - 1935) proved a theorem that became the foundation of modern theoretical physics [9]: *for every continuous symmetry of a physical system's action, there exists a corresponding conserved quantity.* In classical mechanics, spatial translation symmetry yields momentum conservation, while time translation symmetry yields energy conservation.

In quantum field theory, this principle extends to *internal* symmetries, i.e., transformations of the fields themselves rather than the spacetime coordinates. To see how this arises mathematically, consider a general Lagrangian density $\mathcal{L}(\psi, \partial_\mu \psi)$ that depends on a field ψ and its derivative $\partial_\mu \psi$. Suppose we apply an infinitesimal internal transformation to the field:

$$\psi \to \psi' = \psi + \delta\psi \tag{6}$$

Because the transformation is purely internal, the spacetime coordinates x^μ remain unchanged. For this transformation to represent a true symmetry of the system, the variation of the Lagrangian density must vanish, $\delta\mathcal{L} = 0$. Using the chain rule, we can expand this variation as:

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \psi}\delta\psi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)}\delta(\partial_\mu \psi) = 0 \tag{7}$$

Since the variation and the spacetime derivative commute ($\delta(\partial_\mu\psi) = \partial_\mu(\delta\psi)$), we can rewrite the second term. Furthermore, we can substitute the first term using the Euler-Lagrange equation of motion (eq. (3)). Making these substitutions yields:

$$\begin{aligned} \delta\mathcal{L} &= \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \right) \delta\psi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \partial_\mu(\delta\psi) \\ &= \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \delta\psi \right) = 0 \end{aligned} \tag{8}$$

Notice that the expression inside the parenthesis is a four-vector whose spacetime divergence is zero. If we factor out the infinitesimal continuous parameter $\delta\omega$ that generated the transformation ($\delta\psi = \frac{\delta\psi}{\delta\omega}\delta\omega$), we arrive at Noether's first theorem, which mathematically guarantees the existence of a conserved four-current j^μ such that the continuity equation $\partial_\mu j^\mu = 0$ in eq. (8) is satisfied. This conserved current is given by:

$$\boxed{j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \frac{\delta\psi}{\delta\omega}} \tag{9}$$

This current is what we will use to construct the Dirac field Lagrangian in the next section.

IV. GLOBAL $U(1)$ GAUGE TRANSFORMATION

For a classical wave, like a sound wave or a ripple on a pond, the phase tells you exactly where you are in the wave's cycle at a given moment. If the wave is described by $A\cos(kx - \omega t + \phi)$, that ϕ is the phase angle. It shifts the entire wave forward or backward in physical space. If you change the phase, the physical wave moves. In quantum mechanics, matter is described by a field, $\psi(x)$. But unlike a water wave, the "amplitude" of a quantum wave is not a physical distance; it is a complex number with a real and an imaginary part. Because it is a complex number, we can write it in polar form: $\psi(x) = |\psi(x)|e^{i\alpha}$ where α is the phase. But notice where this phase lives: it is an angle on the complex plane. Max Born gave us the rule for how this connects to reality: the physical, measurable universe only cares about the probability of finding a particle, which is the magnitude squared of the wavefunction: $P(x) = |\psi(x)|^2 = \psi^*\psi = (|\psi|e^{-i\theta})(|\psi|e^{i\theta}) = |\psi|^2$. Hence, the phase factor $e^{i\theta}$ completely vanishes. This means that the overall phase of a quantum system is fundamentally unphysical[3]. A phase transformation is simply the act of rotating that unobservable angle. The elegance is the fact that the invariance of the Lagrangian density under these unphysical phase transformations just so happens to produce the theories describing the universe.

We will begin with a global $U(1)$ gauge transformation, which is simply a rotation of our field in the complex space under a constant "phase angle" α. Under this global transformation, $\psi \to \psi' = e^{i\alpha}\psi$, the Lagrangian density (eq. (5)) becomes:

$$\begin{aligned}
\mathcal{L}' &= i\bar{\psi}'\,\gamma^\mu\partial_\mu\psi' - m\bar{\psi}'\psi' \\
&= i(e^{-i\alpha}\bar{\psi})\,\gamma^\mu\partial_\mu(e^{i\alpha}\psi) - m(e^{-i\alpha}\bar{\psi})(e^{i\alpha}\psi) \\
&= (e^{-i\alpha}e^{i\alpha})i\bar{\psi}\,\gamma^\mu\partial_\mu\psi - (e^{-i\alpha}e^{i\alpha})m\bar{\psi}\psi \\
&= i\bar{\psi}\,\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi \\
\mathcal{L}' &= \mathcal{L}
\end{aligned}$$

From Noether's first theorem, this global gauge invariance means there exists a conserved current j^μ such that:

$$\partial_\mu j^\mu = 0 \qquad \text{where} \qquad j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\left(\frac{\delta\psi}{\delta\omega}\right) \tag{10}$$

[3] In other words, you can rotate the "clock face" of the complex number however you want, and the physical universe (the probability distribution) will look exactly the same.

There is only one transformation parameter here (α), so $\omega = \alpha$. Note that $\psi' = e^{i\alpha}\psi$, hence:

$$\begin{aligned} \delta\psi = \psi' - \psi &= e^{i\alpha}\psi - \psi \\ &= (e^{i\alpha} - 1)\psi \\ \delta\psi \approx i\alpha\,\psi \quad &\implies \quad \frac{\delta\psi}{\delta\alpha} = i\psi \end{aligned} \tag{11}$$

where we used $e^{i\alpha} \approx 1 + i\alpha$ in the last line. From the Lagrangian, we have:

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)} = i\bar{\psi}\gamma^\mu \tag{12}$$

Hence, the conserved current is:

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\left(\frac{\delta\psi}{\delta\alpha}\right) = \left(i\bar{\psi}\gamma^\mu\right)(i\psi) \quad \implies \quad \boxed{j^\mu = -\bar{\psi}\gamma^\mu\psi} \tag{13}$$

Before proceeding, it is worth explicitly noting a fundamental mathematical property of this transformation: the $U(1)$ group is *Abelian.* In abstract algebra, an Abelian group is simply one whose elements commute. If we were to apply two successive phase transformations to our Dirac field, the order in which we apply them has no effect on the final state: $e^{i\alpha}e^{i\beta}\psi = e^{i\beta}e^{i\alpha}\psi$. This comes from the fact that our transformation parameter α is just a number, or more formally, a 1×1 matrix. While this commutative property seems mathematically trivial, it has a profound physical consequence. Because the underlying symmetry group is abelian, the resulting gauge field (A_μ) we will see later does not interact with itself.

In contrast, if we were to demand a more complex internal symmetry (such as the "special unitary" $SU(2)$ transformations that govern the weak nuclear force), the transformation parameters would be larger matrices that do not commute. This non-abelian geometry makes things a bit more complicated and mathematically forces the resulting gauge bosons (like the W and Z bosons) to carry charge and interact with each other.

V. LOCAL $U(1)$ GAUGE TRANSFORMATION

Now, let us consider a *local* $U(1)$ gauge transformation. That is, the phase angle or transformation parameter α is now spacetime-dependent: $\psi \to \psi' = e^{i\alpha(x)}\psi$. The Lagrangian density becomes:

$$\begin{aligned} \mathcal{L}' &= i\bar{\psi}'\,\gamma^\mu\partial_\mu\psi' - m\bar{\psi}'\psi' \\ &= i(e^{-i\alpha}\bar{\psi})\,\gamma^\mu\partial_\mu(e^{i\alpha}\psi) - m(e^{-i\alpha}\bar{\psi})(e^{i\alpha}\psi) \end{aligned}$$

$$
\begin{aligned}
&= i(e^{-i\alpha}\bar{\psi})\,\gamma^\mu \left[e^{i\alpha}\partial_\mu\psi + ie^{i\alpha}\psi\partial_\mu\alpha\right] - (e^{-i\alpha}e^{i\alpha})m\bar{\psi}\psi \\
&= \left[i\bar{\psi}\,\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi\right] - \bar{\psi}\gamma^\mu\psi\,\partial_\mu\alpha \\
\mathcal{L}' &= \mathcal{L} - \bar{\psi}\gamma^\mu\psi\,\partial_\mu\alpha
\end{aligned}
$$

Hence $\mathcal{L}' \neq \mathcal{L}$. Evidently, a spacetime-dependent transformation parameter ruins the gauge invariance of the Lagrangian density presented in section IV.

If you have studied General Relativity (GR) through the lens of differential geometry (like in Sean Carroll's *Spacetime and Geometry)*, this exact phenomenon should feel strikingly familiar. In GR, demanding diffeomorphism invariance (that physics shouldn't care about your choice of coordinates) forces us to replace the partial derivative ∂_μ with a spacetime covariant derivative ∇_μ. We introduce the Christoffel connection $\Gamma^\lambda_{\mu\nu}$ specifically to cancel out the non-tensorial "garbage terms" generated by coordinate transformations. It should therefore come as no surprise that we'll do something similar here: we replace the ordinary partial derivative ∂_μ with a *gauge invariant derivative* $\mathcal{D}_\mu$, designed to cancel out the extra terms that arises during a local gauge transformation. This gauge invariant derivative or simply covariant derivative in $U(1)$ gauge theory is given by:

$$\boxed{\mathcal{D}_\mu = \partial_\mu - ieA_\mu(x)} \tag{14}$$

Notice that here, the new field $A_\mu(x)$ takes the same role as the Christoffel connection in GR. As we will check later, local gauge invariance will be achieved provided that $A_\mu(x)$ transforms as:

$$\boxed{A'_\mu(x) = A_\mu(x) + \frac{1}{e}\partial_\mu\alpha(x)} \tag{15}$$

Hence, our new Lagrangian density becomes:

$$\mathcal{L} = i\bar{\psi}\,\gamma^\mu\mathcal{D}_\mu\psi - m\bar{\psi}\psi \tag{16}$$

Under the local gauge transformation, our covariant derivative transforms as:

$$\mathcal{D}'_\mu = \partial_\mu - ieA'_\mu = \partial_\mu - ie\left(A_\mu + \tfrac{1}{e}\partial_\mu\alpha\right) \tag{17}$$

$$= (\partial_\mu - ieA_\mu) - i\partial_\mu\alpha$$

$$\mathcal{D}'_\mu = \mathcal{D}_\mu - i\partial_\mu\alpha \tag{18}$$

Hence:

$$\mathcal{D}'_\mu\psi' = (\mathcal{D}_\mu - i\partial_\mu\alpha)\,e^{i\alpha}\psi = \mathcal{D}_\mu(e^{i\alpha}\psi) - i(\partial_\mu\alpha)e^{i\alpha}\psi$$

$$
\begin{aligned}
&= (\partial_\mu - ieA_\mu)e^{i\alpha}\psi - i(\partial_\mu\alpha)e^{i\alpha}\psi \\
&= \cancel{i(\partial_\mu\alpha)e^{i\alpha}\psi} + e^{i\alpha}\partial_\mu\psi - ieA_\mu e^{i\alpha}\psi - \cancel{i(\partial_\mu\alpha)e^{i\alpha}\psi} \\
&= e^{i\alpha}\left(\partial_\mu\psi - ieA_\mu\right)\psi \\
\mathcal{D}'_\mu\psi' &= e^{i\alpha}\mathcal{D}_\mu\psi
\end{aligned}
$$

The Lagrangian density in eq. (16) then becomes:

$$
\begin{aligned}
\mathcal{L}' &= i\bar{\psi}'\,\gamma^\mu\mathcal{D}'_\mu\psi' - m\bar{\psi}'\psi' \\
&= i(e^{-i\alpha}\bar{\psi})\,\gamma^\mu\left[e^{i\alpha}\mathcal{D}_\mu\psi\right] - m(e^{-i\alpha}\bar{\psi})(e^{i\alpha}\psi) \\
&= i(e^{-i\alpha}\bar{\psi})\,\gamma^\mu\left[e^{i\alpha}\partial_\mu\psi + ie^{i\alpha}\psi\partial_\mu\alpha\right] - (e^{-i\alpha}e^{i\alpha})m\bar{\psi}\psi \\
&= (e^{-i\alpha}e^{i\alpha})i\bar{\psi}\,\gamma^\mu\mathcal{D}_\mu\psi - (e^{-i\alpha}e^{i\alpha})m\bar{\psi}\psi \\
&= i\bar{\psi}\,\gamma^\mu\mathcal{D}_\mu\psi - m\bar{\psi}\psi \\
\mathcal{L}' &= \mathcal{L}
\end{aligned}
$$

Thus, local gauge invariance is achieved through the introduction of a new field $A_\mu(x)$ via the substitution, $\partial_\mu \to \mathcal{D}_\mu$. Let us analyze the contents of the gauge-invariant Lagrangian density (eq. (16)) by writing it out explicitly:

$$
\begin{aligned}
\mathcal{L} &= i\bar{\psi}\,\gamma^\mu\mathcal{D}_\mu\psi - m\bar{\psi}\psi && (19) \\
&= i\bar{\psi}\,\gamma^\mu(\partial_\mu - ieA_\mu)\psi - m\bar{\psi}\psi && (20) \\
&= \left[i\bar{\psi}\,\gamma^\mu\,\partial_\mu\psi - m\bar{\psi}\psi\right] + e\bar{\psi}\gamma^\mu\psi A_\mu && (21)
\end{aligned}
$$

But from eq. (13), the $\bar{\psi}\gamma^\mu\psi$ term is just the negative of the conserved current. Hence:

$$
\mathcal{L} = \left[i\bar{\psi}\,\gamma^\mu\,\partial_\mu\psi - m\bar{\psi}\psi\right] - ej^\mu A_\mu \quad\Longrightarrow\quad \mathcal{L} = \mathcal{L}_{free} + \mathcal{L}_{int} \tag{22}
$$

where $\mathcal{L}_{free} = i\bar{\psi}\,\gamma^\mu\,\partial_\mu\psi - m\bar{\psi}\psi$ and $\mathcal{L}_{int} = -ej^\mu A_\mu$. Hence, the requirement for local gauge invariance forces us to introduce a new field A_μ that interacts with the Dirac current j^μ.

Our next task is to introduce a kinetic term for A_μ into the Lagrangian density. While the dynamics of the Dirac field ψ are already accounted for in $\mathcal{L}_{\text{free}}$, the gauge field A_μ currently lacks a term containing its derivatives. Without a kinetic term to dictate how A_μ propagates and evolves through spacetime, it remains a mere mathematical construct. To transition the gauge field from an abstract symmetry requirement into an independent physical reality, we must grant it its own kinetic energy. Fortunately, we don't have to look far, as the gauge-invariant derivative $\mathcal{D}_\mu$ can be a source of the kinetic term since it contains ∂_μ and was the entry point of the field A_μ.

Let us calculate the commutator of our gauge covariant derivative:

$$
\begin{aligned}
[\mathcal{D}_\mu, \mathcal{D}_\nu]\,\psi &= (\mathcal{D}_\mu\mathcal{D}_\nu - \mathcal{D}_\nu\mathcal{D}_\mu)\psi \\
&= (\partial_\mu - ieA_\mu)(\partial_\nu - ieA_\nu)\psi - (\partial_\nu - ieA_\nu)(\partial_\mu - ieA_\mu)\psi \\
&= (\partial_\mu - ieA_\mu)(\partial_\nu\psi - ieA_\nu\psi) - (\partial_\nu - ieA_\nu)(\partial_\mu\psi - ieA_\mu\psi) \\
&= \left[\cancel{\partial_\mu\partial_\nu\psi} - ie\,\partial_\mu(A_\nu\psi) - ieA_\mu(\partial_\nu\psi) + \cancel{e^2A_\mu A_\nu\psi}\right] \\
&\quad - \left[\cancel{\partial_\nu\partial_\mu\psi} - ie\,\partial_\nu(A_\mu\psi) - ieA_\nu(\partial_\mu\psi) + \cancel{e^2A_\nu A_\mu\psi}\right] \\
&= ie\left[-\cancel{A_\nu\partial_\mu\psi} - (\partial_\mu A_\nu)\psi - \cancel{A_\mu(\partial_\nu\psi)}\right] - ie\left[-\cancel{A_\mu\partial_\nu\psi} - \partial_\nu A_\mu)\psi - \cancel{A_\nu(\partial_\mu\psi)}\right] \\
&= -ie(\partial_\mu A_\nu - \partial_\nu A_\mu)\psi \\
[\mathcal{D}_\mu, \mathcal{D}_\nu]\,\psi &= -ieF_{\mu\nu}\psi
\end{aligned}
\tag{23}
$$

where we have defined:

$$\boxed{F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu} \tag{24}$$

Since eq. (24) is a rank 2 tensor, we can contract the indices by defining:

$$\mathcal{L}_{A_\mu} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{25}$$

Adding this as the kinetic term of the A_μ field completes the pieces of the Lagrangian density, where eq. (22) becomes:

$$\mathcal{L} = \mathcal{L}_{free} + \mathcal{L}_{A_\mu} + \mathcal{L}_{int} \tag{26}$$

$$\mathcal{L} = i\bar{\psi}\,\gamma^\mu\,\partial_\mu\psi - m\bar{\psi}\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - ej^\mu A_\mu \tag{27}$$

VI. THE FARADAY TENSOR AND MAXWELL'S EQUATIONS

What is the physical significance of the tensor $F_{\mu\nu}$? For starters, eq. (24) tells us that the tensor is anti-symmetric, $F_{\mu\nu} = -F_{\nu\mu}$. Consequently, it means that the diagonal terms are zero and $F_{ij} = -F_{ji}$ where $i \neq j$:

$$F_{\mu\nu} = \begin{pmatrix} 0 & F_{01} & F_{02} & F_{03} \\ -F_{01} & 0 & F_{12} & F_{13} \\ -F_{02} & -F_{12} & 0 & F_{23} \\ -F_{03} & -F_{13} & -F_{23} & 0 \end{pmatrix} \tag{28}$$

Let $A^\mu = (A_0, \mathbf{A})$, then $A_\mu = (A_0, -\mathbf{A})$[4]. Recall that $\partial_\mu = \left(\frac{1}{c}\frac{\partial}{\partial t}, \nabla\right)$. Hence:

$$F_{0i} = \partial_0 A_i - \partial_i A_0 \qquad \Longrightarrow \qquad F_{0i} = \frac{1}{c}\frac{\partial A_i}{\partial t} - \partial_i A_0 \tag{29}$$

Because A_i represents the components of $-\mathbf{A}$ (meaning $A_1 = -A_x$, $A_2 = -A_y$, $A_3 = -A_z$), writing this out explicitly for each spatial direction yields:

$$F_{01} = -\frac{1}{c}\frac{\partial A_x}{\partial t} - \frac{\partial A_0}{\partial x}; \qquad F_{02} = -\frac{1}{c}\frac{\partial A_y}{\partial t} - \frac{\partial A_0}{\partial y}; \qquad F_{03} = -\frac{1}{c}\frac{\partial A_z}{\partial t} - \frac{\partial A_0}{\partial z} \tag{30}$$

Let us define a 3D vector field $\tilde{\mathbf{E}} \equiv (\tilde{E}_x, \tilde{E}_y, \tilde{E}_z)$, such that:

$$\tilde{\mathbf{E}} = -\frac{1}{c}\frac{\partial \mathbf{A}}{\partial t} - \nabla A_0 \qquad \text{or} \qquad \tilde{E}^i = -\partial_0 A^i - \partial_i A^0 \tag{31}$$

Using $\tilde{E}_i = -\tilde{E}^i$ and $A_i = -A^i$, we get:

$$-\tilde{E}_i = \partial_0 A_i - \partial_i A_0 \qquad \Longrightarrow \qquad \boxed{F_{0i} = -\tilde{E}_i = \left(\tilde{E}_x, \tilde{E}_y, \tilde{E}_z\right)} \tag{32}$$

Using the anti-symmetry property of the the tensor, we have $\boxed{F_{i0} = \tilde{E}_i = \left(-\tilde{E}_x, -\tilde{E}_y, -\tilde{E}_z\right)}$. For the spatial components, we have:

$$F_{ij} = \partial_i A_j - \partial_j A_i \tag{33}$$

Recall that for any vector field $\mathbf{A}$, the curl in index notation is given by:

$$(\nabla \times \mathbf{A})^i = -\frac{1}{2}\epsilon^{ijk}(\partial_j A_k - \partial_k A_j) \tag{34}$$

where again, the negative sign comes from the fact that in our metric signature $A^i = -A_i$. Let us define a 3D vector field $\tilde{\mathbf{B}}$ such that:

$$\tilde{\mathbf{B}} \equiv \nabla \times \mathbf{A} \tag{35}$$

Then by eq. (34), we have:

$$\tilde{B^i} = (\nabla \times \mathbf{A})^i = -\frac{1}{2}\epsilon^{ijk}(\partial_j A_k - \partial_k A_j) = -\frac{1}{2}\epsilon^{ijk}F_{jk} \tag{36}$$

Multiplying both sides by 2 and ϵ_{imn}, we get:

$$-\epsilon_{imn}\epsilon^{ijk}F_{jk} = 2\epsilon_{imn}\tilde{B^i} \tag{37}$$

[4] Remember that we are using the mostly-negative metric signature $(+,-,-,-)$, so the spatial components flip signs when lowering an index.

Using the "epsilon-killer" identity, $\epsilon_{imn}\epsilon^{ijk} = \delta^j_m\delta^k_n - \delta^k_m\delta^j_n$, we get:

$$-\left(\delta^j_m\delta^k_n - \delta^k_m\delta^j_n\right) F_{jk} = 2\epsilon_{imn}\tilde{B}^i \quad \Longrightarrow \quad F_{nm} - F_{mn} = 2\epsilon_{imn}\tilde{B}^i \tag{38}$$

Using the anti-symmetry property of the tensor and changing the indices, we have:

$$-2F_{mn} = 2\epsilon_{imn}\tilde{B}^i \quad \Longrightarrow \boxed{F_{ij} = -\epsilon_{ijk}\tilde{B}^k} \tag{39}$$

Thus, the tensor $F_{\mu\nu}$ in matrix form (eq. (28)) is given by:

$$F_{\mu\nu} = \begin{pmatrix} 0 & \tilde{E}_x & \tilde{E}_y & \tilde{E}_z \\ -\tilde{E}_x & 0 & -\tilde{B}_z & \tilde{B}_y \\ -\tilde{E}_y & \tilde{B}_z & 0 & -\tilde{B}_x \\ -\tilde{E}_z & -\tilde{B}_y & \tilde{B}_x & 0 \end{pmatrix} \tag{40}$$

The contravariant form $F^{\mu\nu}$ can be obtained by raising the indices using the flat Minkowski metric. Because the spatial components of the metric are negative in our chosen signature, lowering one spatial index flips the sign, and lowering two spatial indices leaves the sign unchanged:

$$F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta} = \eta^{00}\eta^{ij}F_{0j} + \eta^{im}\eta^{jn}F_{mn} = \begin{pmatrix} 0 & -\tilde{E}_x & -\tilde{E}_y & -\tilde{E}_z \\ \tilde{E}_x & 0 & -\tilde{B}_z & \tilde{B}_y \\ \tilde{E}_y & \tilde{B}_z & 0 & -\tilde{B}_x \\ \tilde{E}_z & -\tilde{B}_y & \tilde{B}_x & 0 \end{pmatrix} \tag{41}$$

Notice if we take the curl of $\tilde{\mathbf{E}}$ in eq. (31), we get:

$$\nabla \times \tilde{\mathbf{E}} = \nabla \times \left(-\frac{1}{c}\frac{\partial \mathbf{A}}{\partial t} - \nabla A_0\right) = -\frac{1}{c}\frac{\partial}{\partial t}\left(\nabla \times \mathbf{A}\right) - \nabla \times \left(\nabla A_0\right) \tag{42}$$

Since $\nabla \times \mathbf{A}$ is just $\tilde{\mathbf{B}}$ (eq. (35)) and the curl of a gradient is always zero, we get:

$$\boxed{\nabla \times \tilde{\mathbf{E}} = -\frac{1}{c}\frac{\partial \tilde{\mathbf{B}}}{\partial t}} \tag{43}$$

Meanwhile, if we take the divergence of $\tilde{\mathbf{B}}$ in eq. (35), we get:

$$\nabla \cdot \tilde{\mathbf{B}} = \nabla \cdot (\nabla \times \mathbf{A}) \quad \Longrightarrow \quad \boxed{\nabla \cdot \tilde{\mathbf{B}} = 0} \tag{44}$$

since the divergence of a curl is always zero.

If we use our Lagrangian density eq. (27) with the field A_ν and its derivative $\partial_\mu A_\nu$ as the dynamical variables, we will obtain[5]:

$$\boxed{\partial_\mu F^{\mu\nu} = ej^\nu} \tag{45}$$

The timelike ($\nu = 0$) component of this equation is:

$$\partial_\mu F^{\mu 0} = \partial_0 F^{00} + \partial_i F^{i0} = ej^0 \quad \Longrightarrow \quad \partial_i \left(\tilde{E}^i\right) = ej^0 \quad \Longrightarrow \quad \boxed{\nabla \cdot \tilde{\mathbf{E}} = ej^0} \tag{46}$$

while the spatial ($\nu = i$) components are:

$$\partial_\mu F^{\mu i} = \partial_0 F^{0i} + \partial_j F^{ji} = ej^i \quad \Longrightarrow \quad \partial_0 \left(-\tilde{E}^i\right) + \partial_j \left(\epsilon^{ijk} \tilde{B}_k\right) = ej^i \tag{47}$$

$$\Longrightarrow \quad \epsilon^{ijk} \left(\partial_j \tilde{B}_k\right) = ej^i + \partial_0 \tilde{E}^i \tag{48}$$

$$\Longrightarrow \quad \boxed{\nabla \times \tilde{\mathbf{B}} = e\,\mathbf{j} + \frac{1}{c}\frac{\partial \tilde{\mathbf{E}}}{\partial t}} \tag{49}$$

At this point, the profound nature of what we have just done should sink in. Look closely at eqs. (43), (44), (46) and (49). They are ***Maxwell's equations*** in classical electromagnetism.

We did not start with magnets, coils, or laboratory experiments. We simply took a free Dirac Lagrangian in eq. (5) and demanded that the universe shouldn't care about our choice of local phase (a local $U(1)$ symmetry). To maintain that symmetry, the math *forced* us to introduce the vector field, A_μ. When we analyzed the dynamics of that field, classical electromagnetism naturally fell out. In modern physics, this is the ultimate paradigm: **symmetry dictates dynamics**.

Furthermore, the origin of these four equations is beautifully split. While the two inhomogeneous equations (eqs. (46) and (49)) come directly from the dynamics dictated by the Lagrangian, the two homogeneous equations (eqs. (43) and (44)) actually arise naturally from the structural geometry of the $F_{\mu\nu}$ tensor itself, satisfying a geometric constraint known as the **Bianchi Identity**[6]:

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0 \tag{50}$$

Again, if you have studied General Relativity, you might remember the Bianchi identity that is written in terms of the Riemann curvature tensor. That is because in GR, the Christoffel symbols essentially act as the gauge fields, while the Riemann curvature tensor acts as the field strength tensor, perfectly analogous to $F_{\mu\nu}$. Just as demanding local phase invariance gives you the gauge

[5] A detailed derivation of this is found in Appendix C

[6] A detailed derivation of this is found in Appendix C 1

field A_μ, demanding local coordinate invariance forces the existence of spacetime curvature and gravity.

To fully recover the macroscopic Maxwell's equations in standard SI units, we simply map our gauge theory variables to the familiar classical quantities we all know and love. We relate our fields $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{B}}$ to the standard electric and magnetic fields, our field A_μ to the standard electric and magnetic potentials, and our abstract coupling constant e to the fundamental unit of charge:

- **The Fields:** $\tilde{\mathbf{E}} \to \mathbf{E}/c \quad \text{and} \quad \tilde{\mathbf{B}} \to \mathbf{B}$
- **The Potentials:** $A_0 = V/c \quad \text{and} \quad \mathbf{A} \to \mathbf{A}$
- **The Sources:** $ecj^0 = \rho/\epsilon_0 \quad \text{and} \quad ej^i = \mu_0 J^i$

With these substitutions, our abstract $U(1)$ tensor Faraday tensor takes its final, recognizable form:

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix} \tag{51}$$

This tensor is called the electromagnetic field tensor or simply "Faraday", in honor of the great Michael Faraday (1791 - 1867), whose experimental intuition first mapped the fields we have just derived. While we arrived at the Faraday tensor through the $U(1)$ gauge formalism, one can also deduce the necessity of a rank-2 antisymmetric tensor by generalizing the classical Lorentz force law in a relativistically covariant form. This alternative approach is discussed in Appendix B.

VII. THE FORBIDDEN MASS TERM: WHY THE GAUGE FIELD A_μ IS MASSLESS

A careful reader might notice a glaring asymmetry in our finalized Lagrangian density. The Dirac field ψ possesses a mass term, $-m\bar{\psi}\psi$, which dictates the invariant rest mass of the fermion. However, there is no equivalent mass term for our newly minted gauge field A_μ. Why is that?

In classical field theory, if such a vector field were to possess a mass, its allowable mass term (known as a Proca mass term) would be given by:

$$\mathcal{L}_{\text{mass}} = \frac{1}{2} m_\gamma^2 A_\mu A^\mu \tag{52}$$

where m_γ is the mass of the gauge boson. Let us test what happens to this hypothetical mass term if we apply our local $U(1)$ gauge transformation, $A_\mu \to A'_\mu = A_\mu + \frac{1}{e}\partial_\mu \alpha$:

$$\begin{aligned}
\mathcal{L}'_{\text{mass}} &= \frac{1}{2}m_\gamma^2 A'_\mu A'^\mu \\
&= \frac{1}{2}m_\gamma^2 \left(A_\mu + \frac{1}{e}\partial_\mu \alpha\right)\left(A^\mu + \frac{1}{e}\partial^\mu \alpha\right) \\
&= \frac{1}{2}m_\gamma^2 \left[A_\mu A^\mu + \frac{1}{e}A_\mu(\partial^\mu \alpha) + \frac{1}{e}A^\mu(\partial_\mu \alpha) + \frac{1}{e^2}(\partial_\mu \alpha)(\partial^\mu \alpha)\right]
\end{aligned}$$

Because the transformation parameter $\alpha(x)$ varies across spacetime, its derivatives $\partial_\mu \alpha$ are non-zero. Consequently:

$$\mathcal{L}'_{\text{mass}} \neq \mathcal{L}_{\text{mass}} \tag{53}$$

The inclusion of a mass term explicitly and violently breaks local gauge invariance. The mathematics leaves us with only one possible conclusion: to preserve the local $U(1)$ symmetry that generated electromagnetism in the first place, the mass of the gauge field must be exactly zero ($m_\gamma = 0$).

This is not a mere mathematical artifact; it is a profound prediction about the physical universe. It tells us that the photon (the quantum manifestation of the A_μ field) must be strictly massless. Because it lacks mass, the photon does not experience time and travels at the ultimate speed limit of the universe, c. Furthermore, the lack of a mass term prevents the electromagnetic force from decaying exponentially over short distances, which is precisely why the electromagnetic force has an infinite range.

This requirement also highlights the sheer brilliance of the Higgs mechanism mentioned in the conclusion (section VIII). When physicists later tried to apply this exact gauge architecture to the weak nuclear force, the symmetry demanded massless bosons, but experiments showed the W and Z bosons were incredibly massive. Generating mass without breaking the sacred gauge symmetry required the introduction of an entirely new field called the *Higgs field* to dynamically grant mass to the bosons through spontaneous symmetry breaking [10].

VIII. CONCLUSION

We have demonstrated that the classical electrodynamics is not a random collection of empirical rules, but a strict mathematical necessity arising from a single, elegant postulate: the requirement of local $U(1)$ gauge invariance. By demanding that the quantum phase of the Dirac field can be independently chosen at each point in spacetime, we were forced to introduce the gauge field A_μ, construct the gauge covariant derivative $\mathcal{D}_\mu$, and formulate the kinetic Faraday tensor $F_{\mu\nu}$. The familiar electric and magnetic fields, along with Maxwell's equations, emerge organically as the low-energy, classical manifestations of this underlying quantum symmetry.

This profound realization serves as the blueprint for the entire Standard Model of particle physics. By extending this principle to non-Abelian symmetry groups, physicists successfully modeled the other fundamental forces. The weak nuclear force is dictated by local $SU(2)$ symmetry, and its unification with electromagnetism into the electroweak theory by Abdus Salam, Steven Weinberg, and Sheldon Glashow marks one of the greatest triumphs of 20th-century physics [3, 4]. However, pure gauge symmetry strictly requires its force-carrying bosons to be massless, a glaring contradiction with the massive W and Z bosons observed in weak interactions. This discrepancy was elegantly resolved by the Higgs mechanism [5], which preserves the underlying gauge symmetry while dynamically generating mass. The historic discovery of the Higgs boson by the ATLAS and CMS collaborations at CERN in 2012 [6] provided the definitive confirmation of this framework, proving that nature is, at its deepest level, governed by the beautiful dictates of gauge symmetry.

A careful reader might wonder: if the Higgs mechanism so elegantly solves the mass problem for the weak nuclear force, why don't we apply it to electromagnetism? The answer lies in the fundamental difference between a *broken* and an *unbroken* symmetry. The Higgs mechanism is specifically required when a continuous gauge symmetry is spontaneously broken by the vacuum state of the universe, which dynamically forces the associated gauge bosons to acquire mass. However, the $U(1)$ gauge symmetry of electromagnetism is an exact, unbroken symmetry of nature. The vacuum does not hide or distort this phase invariance. Consequently, the photon remains strictly massless, preserving the inverse-square law and the infinite reach of the electromagnetic force. We do not use the Higgs mechanism for the A_μ field simply because nature does not require it; the photon's masslessness is not a theoretical defect to be fixed, but a fundamental feature of our universe.

[1] R. P. Feynman, F. B. Morinigo, and W. G. Wagner, *Feynman Lectures on Gravitation*, edited by B. Hatfield. Addison-Wesley, Reading, MA, 1995.

[2] S. Weinberg. *Gravitation and cosmology: principles and applications of the general theory of relativity.* New York: Wiley, 1998.

[3] A. Salam, "Weak and Electromagnetic Interactions," in *Elementary Particle Theory: Relativistic Groups and Analyticity*, edited by N. Svartholm (Almqvist & Wiksell, 1968), pp. 367–377.

[4] S. Weinberg, "A Model of Leptons," *Phys. Rev. Lett.* **19**, 1264 (1967).

[5] P. W. Higgs, "Broken Symmetries and the Masses of Gauge Bosons," *Phys. Rev. Lett.* **13**, 508 (1964).

[6] G. Aad *et al.* (ATLAS Collaboration), "Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC," *Phys. Lett. B* **716**, 1–29 (2012).

[7] D. Griffiths. *Introduction to electrodynamics*. Cambridge University Press. 2023.

[8] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (CRC Press, 1995).

[9] E. Noether, "Invariante Variationsprobleme," *Nachr. d. König. Gesellsch. d. Wiss. zu Göttingen, Math-phys. Klasse*, 235–257 (1918). [English translation: M. A. Tavel, *Transp. Theory Stat. Phys.* **1**, 183–207 (1971)].

[10] T. Lancaster and S. J. Blundell, *Quantum Field Theory for the Gifted Amateur*. Oxford University Press, 2014.

Appendix A: Derivation of the Dirac Equation

To address the problems encountered by the Klein-Gordon equation (like the negative probability density), P.A.M Dirac (1902 - 1984) derived a quantum relativistic version of the Schrödinger equation that now bears his name. Dirac searched for a quantum relativistic equation that is first-order in time just like the Schrödinger equation, but is compatible with special relativity.

To derive the Dirac equation, one could start with the typical eigenvalue equation in quantum mechanics:

$$H_D\,\psi = \mathcal{E}\,\psi \tag{A1}$$

wherein we require that the quantum particle must also satisfy the special relativistic equation $\mathcal{E}^2 = \mathbf{p}^2c^2 + m^2c^4$, that is,

$$\left(\mathcal{E}^2 - \mathbf{p}^2c^2 - m^2c^4\right)\psi = 0 \tag{A2}$$

The idea is to to essentially "square root" the relativistic energy-momentum relation to obtain the linear form required by eq. (A1). A natural candidate for H_D is therefore a linearized Hamiltonian proportional to the momentum $\mathbf{p}$ and mass m, such as:

$$H_D = (\boldsymbol{\alpha}\cdot\mathbf{p})c + \beta\,mc^2 \tag{A3}$$

Substituting this to eq. (A1) gives:

$$\left[\mathcal{E} - (\boldsymbol{\alpha}\cdot\mathbf{p})c - \beta\,mc^2\right]\,\psi = 0 \tag{A4}$$

To determine the nature of $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$ and β, we must ensure that eq. (A4) is consistent with special relativity and could reproduce eq. (A2). For convenience, let $\hat{M} = (\boldsymbol{\alpha}\cdot\mathbf{p})c + \beta\,mc^2$, so that eq. (A4) becomes

$$(\mathcal{E} - \hat{M})\psi = 0 \tag{A5}$$

Multiplying $(\mathcal{E} + \hat{M})$ on both sides from the left and noting that $\mathcal{E}$ is just a number and commutes with $\hat{M}$, we get:

$$(\mathcal{E} + \hat{M})(\mathcal{E} - \hat{M})\,\psi = 0 \quad\implies\quad (\mathcal{E}^2 - \hat{M}^2)\,\psi = 0 \tag{A6}$$

Now,

$$\hat{M}^2 = [(\boldsymbol{\alpha}\cdot\mathbf{p})c + \beta\,mc^2][(\boldsymbol{\alpha}\cdot\mathbf{p})c + \beta\,mc^2]$$

$$
\begin{aligned}
&= (\boldsymbol{\alpha}\cdot\mathbf{p})(\boldsymbol{\alpha}\cdot\mathbf{p})c^2 + (\boldsymbol{\alpha}\cdot\mathbf{p})\,\beta\,mc^3 + \beta\,mc^3(\boldsymbol{\alpha}\cdot\mathbf{p}) + \beta^2m^2c^4 \\
&= (\alpha_i\,p_i)(\alpha_j\,p_j)c^2 + (\alpha_i\,p_i)\,\beta\,mc^3 + \beta\,mc^3(\alpha_i\,p_i) + \beta^2m^2c^4 \\
\hat{M}^2 &= (\alpha_i\,p_i)(\alpha_j\,p_j)c^2 + mc^3p_i\,(\alpha_i\,\beta + \beta\,\alpha_i) + \beta^2m^2c^4
\end{aligned}
\tag{A7}
$$

where in the 3rd and last line, repeated index implies summation from 1 to 3. Notice that we had to be careful about the commutation of $\boldsymbol{\alpha}$ and β because we do not know what these objects are when we introduced them in eq. (A3).

Let us analyze the first term by writing explicitly the summation symbol in running the indices i and j. In particular, we split the summation depending on whether $i<j$, or $i>j$, i.e.,

$$
(\alpha_i\,p_i)(\alpha_j\,p_j) = \underbrace{(\alpha_i\,p_i)(\alpha_j\,p_j)}_{\text{for } i<j} + \underbrace{(\alpha_i\,p_i)(\alpha_j\,p_j)}_{\text{for } i>j} + \underbrace{\alpha_i^2\,p_i^2}_{\text{for } i=j}
\tag{A8}
$$

If we interchange the dummy indices in the second term, $i \leftrightarrow j$, we would get:

$$
\begin{aligned}
(\alpha_i\,p_i)(\alpha_j\,p_j) &= \underbrace{(\alpha_i\,p_i)(\alpha_j\,p_j)}_{\text{for } i<j} + \underbrace{(\alpha_j\,p_j)(\alpha_i\,p_i)}_{\text{for } j>i} + \underbrace{\alpha_i^2\,p_i^2}_{\text{for } i=j} \\
&= \underbrace{(\alpha_i\,\alpha_j)\,p_i\,p_j}_{\text{for } i<j} + \underbrace{(\alpha_j\,\alpha_i)\,p_i\,p_j)}_{\text{for } i<j} + \underbrace{\alpha_i^2\,p_i^2}_{\text{for } i=j} \\
(\alpha_i\,p_i)(\alpha_j\,p_j) &= (\alpha_i\,\alpha_j + \alpha_j\,\alpha_i)\,p_i\,p_j + \alpha_i^2\,p_i^2
\end{aligned}
\tag{A9}
$$

Hence, eq. (A7) becomes:

$$
\hat{M}^2 = (\alpha_i\,\alpha_j + \alpha_j\,\alpha_i)\,p_i\,p_j\,c^2 + \alpha_i^2\,p_i^2\,c^2 + m\,p_ic^3(\alpha_i\,\beta + \beta\,\alpha_i) + \beta^2m^2c^4
\tag{A10}
$$

Substituting eq. (A10) into eq. (A6), we get:

$$
\left[\mathcal{E}^2 - (\alpha_i\,\alpha_j + \alpha_j\,\alpha_i)\,p_i\,p_j\,c^2 - \alpha_i^2\,p_i^2\,c^2 - m\,p_ic^3(\alpha_i\,\beta + \beta\,\alpha_i) - \beta^2m^2c^4\right]\,\psi = 0
\tag{A11}
$$

Now, the only way eq. (A11) reduces to eq. (A2) is by imposing the following conditions:

$$
(\alpha_i\,\alpha_j + \alpha_j\,\alpha_i) = 0; \quad \text{for } \; i \neq j
\tag{A12}
$$

$$
\alpha_i^2 = 1
\tag{A13}
$$

$$
(\alpha_i\,\beta + \beta\,\alpha_i) = 0
\tag{A14}
$$

$$
\beta^2 = 1
\tag{A15}
$$

The first two relations can be compactly combined using the Kronecker delta, δ_{ij}, which encapsulates the requirement that these objects must anti-commute for different indices but square to 1:

$$\alpha_i\,\alpha_j + \alpha_j\,\alpha_i = 2\delta_{ij} \tag{A16}$$

If we can find mathematical objects, $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$ and β, which obey eqs. (A14) to (A16), then the equation which satisfies both quantum mechanics and special relativity is given by eq. (A4) or:

$$\left[(\boldsymbol{\alpha}\cdot\mathbf{p})c + \beta\, mc^2\right]\,\psi = \mathcal{E}\,\psi \qquad \text{or} \qquad \boxed{\left[-i\hbar\,(\boldsymbol{\alpha}\cdot\nabla) + \beta\, mc\right]\,\psi = i\hbar\frac{1}{c}\frac{\partial\psi}{\partial t}} \tag{A17}$$

where we divided the equation by c and used the quantum substitutions $\mathbf{p} \rightarrow -i\hbar\nabla$ and $\mathcal{E} \rightarrow i\hbar\frac{\partial}{\partial t}$. Equation (A17) is known as the "Schrödinger-like" form of the Dirac equation.

As it turns out, no set of numbers, 1×1, 2×2, or even 3×3 matrices could satisfy eqs. (A14) to (A16). However, a set of 4×4 matrices does exist that satisfies them. These are referred to as the **Dirac matrices** possessing the required anti-commutation properties, and, hence, showing that eq. (A17) satisfies both quantum mechanics and special relativity. These matrices are of the form,

$$\boldsymbol{\alpha} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \tag{A18}$$

where, $\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$, are the 2×2 Pauli spin matrices given by,

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad ; \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad ; \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{A19}$$

The Pauli spin matrices act on the spin state of an electron and were introduced by Wolfgang Pauli (1900 – 1958). The β, on the other hand, is a 4×4 matrix given by,

$$\beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \tag{A20}$$

where I is a 2×2 identity matrix. Since $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$ and β are 4×4 matrices which should act on ψ, then ψ must be a column vector with four components, i.e.,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}. \tag{A21}$$

The matrix multiplications can then be carried out which generates four simultaneous Dirac equations. Solving the four equations yields an explicit form for the four-component spinor ψ. The components ψ_1 and ψ_2 correspond to positive-energy solutions, representing a particle (like an electron) with spin up and spin down, respectively. Conversely, ψ_3 and ψ_4 correspond to negative-energy solutions, representing an antiparticle (like a positron) with spin up and spin down. Although Dirac's original intention was solely to formulate an equation satisfying both quantum mechanics and special relativity, there were profound, unexpected bonuses: his equation naturally incorporated the quantum spin of a particle and predicted the existence of antimatter.

To put the Dirac equation into a covariant form, let us multiply eq. (A17) by β, use eq. (A15), and move all terms to the left-hand side to get:

$$\left(i\hbar\,\beta\,\frac{1}{c}\frac{\partial}{\partial t}+i\hbar\,\beta\,(\boldsymbol{\alpha}\cdot\nabla)-\beta\,mc\right)\psi=0 \tag{A22}$$

(For convenience, we suppress the identity matrix multiplied to the mass term mc). Recall that in the mostly negative signature, the four-momentum is given by $p_\mu=(E/c,-\mathbf{p})$. Hence via the quantum substitution, we have:

$$p_\mu=\left(i\hbar\frac{1}{c}\frac{\partial}{\partial t},i\hbar\nabla\right)=i\hbar\left(\frac{1}{c}\frac{\partial}{\partial t},\nabla\right)=i\hbar\,\partial_\mu \tag{A23}$$

If we define a new four-vector $\gamma^\mu\equiv(\beta,\beta\,\boldsymbol{\alpha})$, one can notice that eq. (A22) takes the form of a covariant inner product of γ^μ and p_μ. Thus, eq. (A22) can be elegantly rewritten as:

$$\boxed{(\gamma^\mu p_\mu-mc)\;\psi=0}\qquad\text{or}\qquad\boxed{(i\hbar\,\gamma^\mu\,\partial_\mu-mc)\;\psi=0} \tag{A24}$$

which is the covariant form of the Dirac equation. As established, γ^μ are the Dirac matrices which obey the fundamental anti-commutation relation (the Clifford algebra):

$$\gamma^\mu\gamma^\nu+\gamma^\nu\gamma^\mu=2\eta^{\mu\nu}I \tag{A25}$$

which is simply based on the relations in eqs. (A14) to (A16). In natural units ($c=\hbar=1$), eq. (A24) takes the form shown in eq. (4), which is the more familiar form found in textbooks.

Appendix B: Alternative Derivation of the Faraday Tensor $F_{\mu\nu}$

In the main text, we arrived at the Faraday tensor $F_{\mu\nu}$ from symmetry principles. But historically and pedagogically, one can also deduce the necessity of a rank-2 antisymmetric tensor purely from the bottom up by generalizing the classical Lorentz force law in a relativistically covariant form.

We begin with the familiar Lorentz force equation for a particle with charge q and momentum $\mathbf{p}$. In vector and index notation, this is:

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = q\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right) \quad \Longrightarrow \quad \frac{dp^i}{dt} = qE^i + \epsilon^{ijk} v_j B_k \tag{B1}$$

To make this equation compatible with special relativity, we must express the derivative with respect to the Lorentz-invariant proper time τ instead of the observer-dependent coordinate time t. We can accomplish this using the chain rule and the time dilation relation $dt = \gamma d\tau$ (where γ is the Lorentz factor):

$$\frac{dp^i}{d\tau}\frac{d\tau}{dt} = qE^i + \epsilon^{ijk} v_j B_k \quad \Longrightarrow \quad \frac{dp^i}{d\tau} = q\gamma E^i + \epsilon^{ijk}(\gamma v_j) B_k \tag{B2}$$

To write this in terms of four-vectors, recall that the four-velocity U^μ is defined as $U^\mu = (c\gamma, \gamma\mathbf{v})$. In the mostly-negative metric signature, the covariant four-velocity is $U_\mu = (c\gamma, -\gamma\mathbf{v})$. Substituting $\gamma = U_0/c$ and $\gamma v_j = -U_j$ into eq. (B2) yields:

$$\frac{dp^i}{d\tau} = qU_0\left(E^i/c\right) - q\epsilon^{ijk}\, U_j B_k \tag{B3}$$

To complete the relativistic picture, we must also find the time component of the momentum derivative, $dp^0/d\tau$, which corresponds to the rate of change of the particle's relativistic energy $\mathcal{E}$ (since $p^0 = \mathcal{E}/c$). Recall that the power P delivered to a particle is the scalar product of force and velocity. Because the magnetic force is always perpendicular to velocity, it does no work; only the electric field contributes. Hence:

$$\begin{aligned} P = \frac{d\mathcal{E}}{dt} = \mathbf{F}\cdot\mathbf{v} \quad \Longrightarrow \quad \frac{d\mathcal{E}}{d\tau}\frac{d\tau}{dt} = qE^i v_i \quad &\Longrightarrow \quad \frac{1}{c}\frac{d\mathcal{E}}{d\tau} = q(\gamma v^i)(E^i/c) \\ &\Longrightarrow \quad \frac{dp^0}{d\tau} = -qU_i(E^i/c) \end{aligned} \tag{B4}$$

To complete the picture, we must combine eq. (B3) and eq. (B4) into a single tensor equation of the form $dp^\mu/d\tau = K^\mu$ where K^μ is the force. Because the electromagnetic force depends linearly on the particle's velocity, the most natural relativistic generalization is to contract the four-velocity

U_ν with a rank-2 tensor $F^{\mu\nu}$ containing the fields:

$$\frac{dp^\mu}{d\tau} = q\,U_\nu F^{\mu\nu} \tag{B5}$$

By demanding that this covariant ansatz reproduces the classical Lorentz force law, we can solve for the components of $F^{\mu\nu}$. For the $\mu = 0$ component, we expand the sum and compare it to our energy equation (eq. (B4)):

$$\frac{dp^0}{d\tau} = q\,U_\nu F^{0\nu} = q\,U_0 F^{00} + q\,U_i F^{0i} = -qU_i(E^i/c) \tag{B6}$$

Hence $F^{00} = 0$ and $F^{0i} = -E^i/c$. Similarly, we expand the spatial components ($\mu = i$) and compare them to our spatial momentum equation (eq. (B3)):

$$\frac{dp^i}{d\tau} = q\,U_\nu F^{i\nu} = q\,U_0 F^{i0} + q\,U_j F^{ij} = q\,U_0(E^i/c) - q\,\epsilon^{ijk} U_j B_k \tag{B7}$$

Hence $F^{i0} = E^i/c$ and $F^{ij} = -\epsilon^{ijk} B_k$. Notice that the diagonal elements are zero ($F^{00} = F^{11} = F^{22} = F^{33} = 0$), and the off-diagonal elements are antisymmetric ($F^{\mu\nu} = -F^{\nu\mu}$). Assembling these components into a 4×4 matrix directly reveals the contravariant electromagnetic field tensor:

$$F^{\mu\nu} = \begin{pmatrix} F^{00} & F^{01} & F^{02} & F^{03} \\ F^{10} & F^{11} & F^{12} & F^{13} \\ F^{20} & F^{21} & F^{22} & F^{23} \\ F^{30} & F^{31} & F^{32} & F^{33} \end{pmatrix} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix} \tag{B8}$$

The covariant form $F_{\mu\nu}$ can be obtained by lowering the indices using the flat Minkowski metric:

$$F_{\mu\nu} = \eta_{\mu\alpha}\eta_{\nu\beta}F^{\alpha\beta} = \eta_{00}\eta_{ij}F^{0j} + \eta_{im}\eta_{jn}F^{mn} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix} \tag{B9}$$

This perfectly matches the Faraday tensor derived via the gauge principle in the main text, showing that classical electrodynamics and local $U(1)$ gauge symmetry are two sides of the exact same mathematical coin.

Appendix C: Maxwell's Equations in Covariant Form

One way to derive Maxwell's equations in covariant form is to begin with the Lagrangian density in eq. (27), taking A_ν and its derivative $\partial_\mu A_\nu$ as the independent field variables. Since the free part of the Dirac Lagrangian density is independent of A_μ, we only need:

$$\mathcal{L} = \mathcal{L}_{A_\mu} + \mathcal{L}_{\text{int}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - ej^\mu A_\mu \tag{C1}$$

The equations of motion can then be obtained via the following Euler-Lagrange equation:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} \right) = \frac{\partial \mathcal{L}}{\partial A_\nu} \tag{C2}$$

We will evaluate the right-hand side and left-hand side independently. For the right-hand side, we take the derivative of the Lagrangian with respect to the field A_ν. Since the free-field term only contains derivatives of the field, it vanishes under this operation:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial A_\nu} &= \frac{\partial}{\partial A_\nu}\left[-\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta} - ej^\alpha A_\alpha\right] \\
&= \frac{\partial}{\partial A_\nu}\left[-\frac{1}{4}(\partial_\alpha A_\beta - \partial_\beta A_\alpha)(\partial^\alpha A^\beta - \partial^\beta A^\alpha) - ej^\alpha A_\alpha\right] \\
&= -ej^\alpha \frac{\partial A_\alpha}{\partial A_\nu} \\
&= -ej^\alpha \, \delta^\nu_\alpha
\end{aligned}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial A_\nu} = -ej^\nu} \tag{C3}$$

This result simply extracts the source current, acting as the driving force for the fields.

For the left-hand side, we must differentiate the kinetic term with respect to $\partial_\mu A_\nu$. To keep track of the indices properly, we write $F_{\alpha\beta}F^{\alpha\beta} = \eta^{\alpha\gamma}\eta^{\beta\delta}F_{\alpha\beta}F_{\gamma\delta}$. Thus:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} &= \frac{\partial}{\partial(\partial_\mu A_\nu)}\left[-\frac{1}{4}\eta^{\alpha\gamma}\eta^{\beta\delta}F_{\alpha\beta}F_{\gamma\delta} - ej^\alpha A_\alpha\right] \\
&= \frac{\partial}{\partial(\partial_\mu A_\nu)}\left[-\frac{1}{4}\eta^{\alpha\gamma}\eta^{\beta\delta}(\partial_\alpha A_\beta - \partial_\beta A_\alpha)(\partial_\gamma A_\delta - \partial_\delta A_\gamma)\right]
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -\frac{1}{4}\eta^{\alpha\gamma}\eta^{\beta\delta}\Bigg[&\left(\frac{\partial(\partial_\alpha A_\beta)}{\partial(\partial_\mu A_\nu)} - \frac{\partial(\partial_\beta A_\alpha)}{\partial(\partial_\mu A_\nu)} \right)(\partial_\gamma A_\delta - \partial_\delta A_\gamma) \\
&+ (\partial_\alpha A_\beta - \partial_\beta A_\alpha)\left(\frac{\partial(\partial_\gamma A_\delta)}{\partial(\partial_\mu A_\nu)} - \frac{\partial(\partial_\delta A_\gamma)}{\partial(\partial_\mu A_\nu)} \right) \Bigg]
\end{aligned}$$

Note that for the derivatives to be non-zero, the indices of the derivatives and the fields must exactly match. This relationship is expressed using Kronecker deltas:

$$\frac{\partial(\partial_\alpha A_\beta)}{\partial(\partial_\mu A_\nu)} = \delta^\mu_\alpha \delta^\nu_\beta \tag{C4}$$

Substituting this back in allows us to evaluate the derivative. Using the the metric tensors to raise the indices and group the resulting terms, we get:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} &= -\frac{1}{4}\eta^{\alpha\gamma}\eta^{\beta\delta}\left[\left(\delta^\mu_\alpha\delta^\nu_\beta - \delta^\mu_\beta\delta^\nu_\alpha\right)(\partial_\gamma A_\delta - \partial_\delta A_\gamma) + (\partial_\alpha A_\beta - \partial_\beta A_\alpha)\left(\delta^\mu_\gamma\delta^\nu_\delta - \delta^\mu_\delta\delta^\nu_\gamma\right)\right] \\
&= -\frac{1}{4}\left[\left(\eta^{\mu\gamma}\eta^{\nu\delta} - \eta^{\nu\gamma}\eta^{\mu\delta}\right)(\partial_\gamma A_\delta - \partial_\delta A_\gamma) + (\partial_\alpha A_\beta - \partial_\beta A_\alpha)\left(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\beta\mu}\eta^{\alpha\nu}\right)\right] \\
&= -\frac{1}{4}\left[\left(\eta^{\mu\gamma}\eta^{\nu\delta} - \eta^{\nu\gamma}\eta^{\mu\delta}\right)F_{\gamma\delta} + \left(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\beta\mu}\eta^{\alpha\nu}\right)F_{\alpha\beta}\right] \\
&= -\frac{1}{4}\left[(F^{\mu\nu} - F^{\nu\mu}) + (F^{\mu\nu} - F^{\nu\mu})\right] \\
&= -\frac{1}{4}\left[F^{\mu\nu} + F^{\mu\nu} + F^{\mu\nu} + F^{\mu\nu}\right] \quad \text{(using the antisymmetry property } F^{\nu\mu} = -F^{\mu\nu}\text{)} \\
&= -\frac{1}{4}\left[4F^{\mu\nu}\right]
\end{aligned}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -F^{\mu\nu}} \tag{C5}$$

Substituting eqs. (C3) and (C5) into Euler-Lagrange equation (eq. (C2)), we get:

$$\partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)}\right) = \frac{\partial \mathcal{L}}{\partial A_\nu} \quad \Longrightarrow \quad \partial_\mu\left(-F^{\mu\nu}\right) = -ej^\nu \quad \Longrightarrow \quad \boxed{\partial_\mu F^{\mu\nu} = ej^\nu} \tag{C6}$$

As shown in the main text, eq. (C6) encapsulates the two inhomogeneous Maxwell's equations: Gauss's law for electricity and Ampere's law.

1. The Homogeneous Equations (Bianchi Identity)

What about the remaining equations, Gauss's law for magnetism and Faraday's law? These do not arise dynamically from the Lagrangian's equations of motion, but rather geometrically from the definition of the field strength tensor itself.

In abstract algebra, the commutator of any operators $\hat{A}$, $\hat{B}$, and $\hat{C}$ must satisfy the Jacobi identity:

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0 \tag{C7}$$

For the gauge covariant derivative $\mathcal{D}_\mu$ (eq. (14)), we have :

$$[D_\mu, [D_\nu, D_\lambda]] + [D_\nu, [D_\lambda, D_\mu]] + [D_\lambda, [D_\mu, D_\nu]] = 0 \tag{C8}$$

The commutator of the covariant derivatives was derived in the main text and is given by eq. (23). Plugging this into eq. (C8), we get:

$$\begin{aligned} [D_\mu, (-ieF_{\nu\lambda})] + [D_\nu, (-ieF_{\lambda\mu})] + [D_\lambda, (-ieF_{\mu\nu})] = 0 \\ [D_\mu, F_{\nu\lambda}] + [D_\nu, F_{\lambda\mu}] + [D_\lambda, F_{\mu\nu}] = 0 \end{aligned} \tag{C9}$$

Because electromagnetism is governed by a $U(1)$ gauge symmetry, which is an Abelian (commutative) gauge theory, the gauge field does not carry charge and therefore does not interact with itself (unlike gluons in quantum chromodynamics). Consequently, when the gauge covariant derivative D_μ acts on the gauge-invariant field strength $F_{\mu\nu}$, it simply reduces to the standard partial derivative ∂_λ.

Thus, the Jacobi identity simplifies to the Bianchi identity for the electromagnetic field tensor:

$$\boxed{\partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0} \tag{C10}$$

This single tensor equation contains the two homogeneous Maxwell's equations: Gauss's law for magnetism and Faraday's law.